Table 1: Configurations of the five evaluation scenarios.

| **Setting** | **S0** | **S1** | **S2** | **S3** | **S4** |
|---|---|---|---|---|---|
| Non-zero initial $\boldsymbol{\omega}_0$ | — | — | ✓ | — | — |
| Measurement noise | — | ✓ | ✓ | ✓ | ✓ |
| Parameter mismatch | — | — | — | ✓ | ✓ |
| Sinusoidal disturbance | — | — | — | ✓ | ✓ |
| Actuator faults | — | — | — | — | ✓ |

**Notes:**

- **Initial conditions:** All scenarios share the same initial attitude $\boldsymbol{p}_0 = [0.3,\ 0.1,\ -0.2]$ and controller gains ($k_p = 30$, $k_d = 60$). S2 uses a non-zero initial angular velocity $\boldsymbol{\omega}_0 = [0.02,\ -0.03,\ 0.04]\,\mathrm{rad/s}$; all other scenarios start from rest ($\boldsymbol{\omega}_0 = \mathbf{0}$).
- Measurement noise: Gaussian white noise on MRP ($\sigma_p = 7.0 \times 10^{-5}$, equivalent to $\sim$15 arcsec star tracker accuracy) and angular velocity ($\sigma_\omega = 5.0 \times 10^{-5}\,\mathrm{rad/s}$, typical MEMS gyro drift level).
- Parameter mismatch: the true moment of inertia matrix is perturbed to $\boldsymbol{J}_{\text{true}} = \begin{bmatrix} 120 & 6 & 7 \\ 6 & 100 & 10 \\ 7 & 10 & 120 \end{bmatrix} \mathrm{kg\,m^2}$ (off-diagonal terms deviate from nominal $\boldsymbol{J}_{\text{nom}} = \mathrm{diag}(110, 100, 120)\,\mathrm{kg\,m^2}$ by up to $10\,\mathrm{kg\,m^2}$), and the flexible parameters $\boldsymbol{\delta}$, $\boldsymbol{\omega}_n$, $\boldsymbol{\xi}$ are all scaled by +10% relative to nominal values.
- Sinusoidal disturbance: $\boldsymbol{d}(t) = [0.02\sin(t+1.2),\ 0.03\sin(t+4.5),\ 0.05\sin(t+0.9)]^\top\,\mathrm{Nm}$. The amplitudes and phases are fixed across evaluation episodes to ensure reproducibility.
- Actuator faults (S4 only): combined multiplicative efficiency loss and additive bias fault modeled as $\boldsymbol{\tau}_{\text{actual}} = (\boldsymbol{I} - \boldsymbol{E})\boldsymbol{\tau}_{\text{cmd}} + \bar{\boldsymbol{u}}$, where the efficiency loss matrix $\boldsymbol{E}(t)$ and bias vector $\bar{\boldsymbol{u}}(t)$ are time-varying piecewise-constant profiles active during $t \in [20, 180]\,\mathrm{s}$.
- These five scenarios form a progressive uncertainty ladder: S0 establishes the pure baseline; S1 adds sensor imperfection; S2 further introduces a non-zero initial angular velocity; S3 layers model-plant mismatch and external sinusoidal disturbance; S4 represents the most demanding condition with concurrent actuator faults.